layouts from chapter 5. A network doesn't have to be a social network. This format can represent many different structures, such as transportation networks and linked open data. In this chapter we'll look at four common forms for representing networks: as data, as adjacency matrices, as arc diagrams, and using force-directed network diagrams.

In each case, the graphical representation will be quite different. For instance, in the case of a force-directed layout, we'll represent the nodes as circles and the edges as lines. But in the case of the adjacency matrix, nodes will be positioned on x- and y-axes and the edges will be filled squares. Networks don't have a default representation, but the examples you'll see in this chapter are the most common.

6.1.1 Network data

Although you can store networks in several data formats, the most straightforward is known as the *edge list*. An edge list is typically represented as a CSV like that shown in listing 6.1, with a source column and a target column, and a string or number to indicate which nodes are connected. Each edge may also have other attributes, indicating the type of connection or its strength, the time period when the connection is valid, its color, or any other information you want to store about a connection. The important thing is that only the source and target columns are necessary.

In the case of directed networks, the source and target columns indicate the direction of connection between nodes. A directed network means that nodes may be connected in one direction but not in the other. For instance, you could follow a user on Twitter, but that doesn't necessarily mean that the user follows you. Undirected networks still typically have the columns listed as "source" and "target," but the connection is the same in both directions. Take the example of a network made up of connections indicating people have shared classes. Then if I'm in a class with you, you're likewise in a class with me. You'll see directed and weighted networks represented throughout this chapter.

Listing 6.1 edgelist.csv

```
source,target,weight
sam,pris,1
roy,pris,5
roy,sam,1
tully,pris,5
tully,kim,3
tully,pat,1
tully,mo,3
kim,pat,2
kim,mo,1
mo,tully,7
mo,pat,1
mo,pris,1
pat,tully,1
pat,kim,2
pat,mo,5
lee,al,3
```